6TH GRADE MATH · GEOMETRY

Parallelogram & Triangle Area Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Parallelogram & Triangle Area Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A rectangle is 6 by 4. You push the top sideways to make a parallelogram, keeping the same base and height. Does the area change?

2. A parallelogram has base 8, height 5 and slanted sides 6. What is its area?

2 Explore use the tool

1. Set base 6 and height 4. Lean the top to 3, then to −2. Record the area each time.

2. With lean 3, press Show cut & slide. What shape do the pieces make, and what are its dimensions?

3. Set base 5, height 4, lean 6. Compare base × slanted side with the real area.

4. Switch to Triangle and show the copy. What shape do the triangle and its copy make? How does that explain the formula?

3 Record use the tool and fill in the table

Areas

Use the sliders. For the last row, find the missing base.

Shape	Base	Height	Lean	Area (square units)
Parallelogram	8	3	0	
Parallelogram	8	3	5	
Triangle	8	3	2	
Triangle	8	3	−3	
Triangle		4	any	12

4 Explain what did you discover?

1. Leaning a parallelogram does not change its _____ or its _____, so the area stays the same.
2. Area of a parallelogram = _____.
3. A triangle is _____ of a parallelogram with the same base and height, so its area = _____.
4. The height is measured _____ from the base, even when it falls outside the shape.
5. Why doesn't the length of the slanted side matter for area?

5 Apply a new situation, no tool

1. Find the area of a parallelogram with base 12 cm, height 7 cm and slanted sides 9 cm.

2. A triangle has an area of 30 in^2 and a base of 10 in. Find its height.

3. Two triangles have the same base and the same height, but one is obtuse. Compare their areas.

4. A triangle's base is doubled and its height stays the same. What happens to the area?